

torch.tile#

<https://pytorch.org/docs/stable/generated/torch.tile.html#torch.tile>

Description:

Constructs a tensor by repeating the elements of input. The dims argument specifies the number of repetitions in each dimension. If dims specifies fewer dimensions than input has, then ones are prepended to dims until all dimensions are specified. For example, if input has shape (8, 6, 4, 2) and dims is (2, 2), then dims is treated as (1, 1, 2, 2). Analogously, if input has fewer dimensions than dims specifies, then input is treated as if it were unsqueezed at dimension zero until it has as many dimensions as dims specifies. For example, if input has shape (4, 2) and dims is (3, 3, 2, 2), then input is treated as if it had the shape (1, 1, 4, 2). This function is similar to NumPy's tile function.

Function Signature

`torch.tile(input, dims) → Tensor#`

Parameters

Code Examples

```
>>> x = torch.tensor([1, 2, 3])
>>> x.tile((2,))
tensor([1, 2, 3, 1, 2, 3])
>>> y = torch.tensor([[1, 2], [3, 4]])
>>> torch.tile(y, (2, 2))
tensor([[1, 2, 1, 2],
        [3, 4, 3, 4],
        [1, 2, 1, 2],
        [3, 4, 3, 4]])
```

Notes & Warnings

NOTE This function is similar to NumPy's `tile` function.

Detailed Documentation

Documentation

Constructs a tensor by repeating the elements of input. The `dims` argument specifies the number of repetitions in each dimension.

If `dims` specifies fewer dimensions than input has, then ones are prepended to `dims` until all dimensions are specified. For example, if input has shape `(8, 6, 4, 2)` and `dims` is `(2, 2)`, then `dims` is treated as `(1, 1, 2, 2)`.

Analogously, if input has fewer dimensions than `dims` specifies, then input is treated as if it were unsqueezed at dimension zero until it has as many dimensions as `dims` specifies. For example, if input has shape `(4, 2)` and `dims` is `(3, 3, 2, 2)`, then input is treated as if it had the shape `(1, 1, 4, 2)`.

This function is similar to NumPy's `tile` function.

`input` (Tensor) – the tensor whose elements to repeat.

`dims` (tuple) – the number of repetitions per dimension.